Grounding Requirements for Portable, Vehicle or Trailer Mounted Generators

Purpose: This handout is to address questions relating to grounding of portable, vehicle and trailer mounted generators, other electrical equipment switch gear, transformers, etc.

NEC 250.34.A Portable Generators
The frame of a portable generator shall not be required to be connected to the grounding electrode as defined in 250.52 for a system supplied by the generator under **both of the following conditions**:

1. The generator supplies only equipment mounted on the generator, cord and plug connected equipment through receptacles mounted on the generator, or both.

2. The normally non-current carrying metal parts of equipment and the equipment grounding conductor terminals of the receptacles are connected to the generator frame.

NEC 250.34.B Vehicle Mounted or Trailer Mounted Generators
The frame of a vehicle or trailer shall not be required to be connected to a grounding electrode as defined in 250.52 for a system supplied by a generator located on this vehicle or trailer under **all the following conditions:**

1. The frame of the generator is bonded to the vehicle or trailer frame

2. The generator supplies only equipment located on the vehicle or trailer; cord and plug connected equipment through receptacles mounted on the vehicle; or both equipment located on the vehicle or trailer and cord and plug connected equipment through receptacles mounted on the vehicle, trailer, or on the generator.

3. The normally non-current-carrying metal parts of equipment and the equipment grounding conductor terminals of the receptacles are connected to the generator frame.

NEC 250.50 Grounding Electrode System. If the portable, vehicle or trailer generator does not meet the conditions above then a grounding electrode system shall be required.

NEC 250-52.A.5 Rod or Pipe Electrodes shall not be less than 2.44 m (8 feet) in length and consist of the following materials.

1. Grounding electrodes of pipe or conduit shall not be smaller than metric designator 21 (trade size ¾ inch) and, where of steel, shall have the outer surface galvanized or otherwise metal coated for corrosion protection.

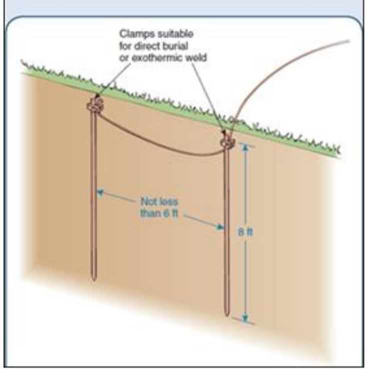
2. Rod type, grounding electrodes of stainless steel and copper or zinc coated steel shall be at least 15.87 mm (5/8 inches) in diameter, unless listed.

3. Other electrodes with BIO Approval.

F&O generators only: If connection to the generator output is by camlock type connectors, and the generator is adjacent to an existing building and with an existing grounding electrode system, then no additional grounding electrodes are required.

If any connection to the generator output are bolted connections, then 2 interconnected ground rods separated by at least six feet are required. The ground rod jumper connectors must be listed for underground installation. Connected by #6 wire or smaller depending on generator.

The spacing requirement in 250.53(A)(3) for multiple auxiliary rod, pipe, or plate electrodes is illustrated in Exhibit 250.24. The direct-buried clamps shown in Exhibit 250.24 are required to be listed for underground installation per 250.70.



Electrical Equipment associated with the generator shall also be grounded for NEC 250.50 ie Switchgear, transformers, etc. NOT spider boxes or distribution boxes.

In addition, all generators need Environmental Protection approval before use. Generators over 50 hp require a Bay Area Air Quality District (BAAQMD) permit. Most generators are approved for 1 year. Additional permits are required over one year.